ML-Based login attack detection and database forensics System

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ABSTRACT

The rapid expansion of web-based applications has increased the risk of cyberattacks targeting authentication systems and databases. This project, titled **"ML-Based Login Attack Detection and Database Forensics System"**, aims to enhance login security by detecting malicious activities such as SQL Injection and Cross-Site Scripting (XSS) attacks using Machine Learning techniques.

The system utilizes a supervised machine learning model trained on datasets containing normal and attack payloads. Text inputs from login credentials are transformed using TF-IDF vectorization and classified using the Random Forest algorithm to identify suspicious or normal behavior. In addition to machine learning-based detection, rule-based filtering is implemented to improve accuracy and real-time attack recognition.

The application is developed using the Flask framework with MySQL database integration for secure storage of login logs. The system records username, hashed password, IP address, timestamp, attack type, and confidence score for forensic analysis. The command prompt also displays logging information for monitoring suspicious activities.

The proposed system helps improve authentication security by providing early attack detection, database forensic support, and automated logging mechanisms, making web applications more resilient against common injection-based attacks.

1.INTRODUCTION

With the rapid growth of internet technology and web applications, security has become a top concern in modern computing systems. Authentication systems like login pages are widely used in banking, social networking, e-commerce, and organizational platforms. However, cyber attackers often target these systems to exploit weaknesses and gain unauthorized access to sensitive data.Among various web security threats, SQL Injection (SQLi) and Cross-Site Scripting (XSS) attacks are particularly dangerous. SQL Injection involves attackers inserting harmful SQL queries into input fields to control backend databases, bypass authentication, or extract confidential information. XSS attacks allow attackers to inject malicious scripts into web pages, which can run in the victim's browser to steal session tokens, credentials, or carry out unauthorized actions.

Traditional security methods mainly depend on rule-based validation and firewall filtering, which often fail to detect complex attack patterns. To address this gap, more security approaches based on machine learning are being adopted. Machine learning models can learn attack patterns from data and automatically classify user inputs as normal or harmful.

This project presents an **ML-Based Login Attack Detection and Database Forensics System** designed to improve authentication security using machine learning techniques. The system uses datasets containing normal and attack payloads to train a classification model. It converts text input from login credentials into numerical features using TF-IDF vectorization, and a Random Forest classifier predicts whether the input is safe or suspicious.

Besides machine learning detection, the system also uses rule-based filtering to boost detection accuracy. When a login attempt occurs, the system examines the username and password fields, identifies possible attack patterns, and logs the event details. The log data includes the username, IP address, timestamp, attack type, and confidence score, all stored in a MySQL database for forensic analysis.

The application is built using the Flask web framework, which provides a lightweight and flexible environment for creating secure web applications. The system also shows detection results in the command prompt for real-time monitoring.

The main goal of this project is to strengthen login security, enable early attack detection, and assist in database forensic investigations .

1.1 ORGANIZATION PROFILE

PSGR Krishnammal College for Women, established in 1963, is a prestigious institution in Coimbatore, Tamil Nadu, dedicated to providing high-quality education for women. Affiliated with Bharathiar University, the college has gained a strong reputation for academic excellence, research, and innovation. It is accredited with an 'A++' grade by NAAC and is recognized as one of the top autonomous women's colleges in India.

The institution offers a diverse range of undergraduate, postgraduate, and doctoral programs in fields such as Science, Commerce, Arts, and Management. With a focus on holistic development, the college integrates modern teaching methodologies, well-equipped laboratories, and digital learning platforms to enhance students' academic experience. The curriculum is designed to be industry-relevant, promoting skill development, entrepreneurship, and leadership qualities among the students.

Through its commitment to women's empowerment, PSGR Krishnammal College continues to shape future leaders by instilling confidence, knowledge, and a spirit of service among its students. The institution stands as a beacon of excellence in higher education, fostering academic brilliance and personal development for women aspiring to make a difference in various professional fields.

1.2 PROBLEM IDENTIFICATION

With the rapid adoption of web-based authentication systems, security vulnerabilities in login modules have become a major concern. Many web applications still rely on traditional validation methods, which are unable to detect advanced injection-based attacks. The major problems identified in existing login security systems are as follows:

a. Lack Of Intelligent Attack Detection

Traditional security mechanisms use simple input validation techniques that cannot detect complex attack patterns such as SQL Injection and Cross-Site Scripting (XSS). Attackers can modify payload structures to bypass static security rules.

b. Absence Of Machine Learning Basic Security Analysis

Most authentication systems do not incorporate machine learning models for behavioral or payload-based threat detection. As a result, unknown or new attack patterns may go undetected.

c.Database Vulnerability Risks

Sensitive login information may be exposed if proper hashing and logging mechanisms are not implemented. Unauthorized database access can lead to data leakage and forensic challenges.

d. Lack Of Real-Time Monitoring

Existing systems rarely provide real-time detection and logging of suspicious activities. Security administrators may not receive immediate alerts when an attack occurs.

e. Difficulty In Attack Forensics Analysis

When security breaches occur, it is difficult to trace attack sources without structured logging of IP address, timestamp, and attack type.

f. Low Accuracy In Manual Security Filtering

Rule-based filtering alone cannot guarantee high detection accuracy, especially against evolving cyberattack techniques.

2. LITERATURE SURVEY

S. No	Title	Author	Focus and objective	Conclusion
1	Five stages of database forensic analysis(2010)	Chopade, Rupali, and Vinod Keshaorao Pachghare	Information Security for South Africa (ISSA)	Explained the five major stages of database forensic investigation in detail.
2	InnoDB database forensics(2011)	Fruhwirt, Peter, et al. "InnoDB database forensics.	Proceedings of 2011 International Conference on Advancements	Focused on forensic analysis of MySQL InnoDB storage engine
3	The role of triggers in database forensics.(2011)	Hauger, Werner K., and Martin S. Olivier	8th Biennial Conference on Innovative	Explained how database triggers help in evidence collection.
4	Ideal log setting for database forensics reconstruction(2013)	Adedayo, Oluwasola Mary, and Martin S. Olivier.	IEEE Access 8 : 112846-112858	Discussed log configuration for accurate forensic reconstruction.

S. No	Title	Author	Focus/Objective	Conclusion
5	On Dimensions of Reconstruction in Database Forensics.(2014)	Fasan, Oluwasola Mary, and Martin S. Olivier.	ICCUBEA -2014	Explained reconstruction techniques using database logs.
6	The state of database forensic research(2014)	Werner K. Hauger, Martin S. Olivier	Applied Cybersecurity & Internet Governance	Gave a complete overview of database forensic research trends.
7	Database security threats and challenges in database forensic:A survey(2015)	Harmeet Kaur Khanuja, D. S. Adane	Journal of Artificial Intelligence and Engineering Applications (JAIEA)	Explained database threats and forensic challenges
8	Database forensic analysis with DBCarver(2016)	James Wagner et al.	Journal of Applied Informatics and Computing	Introduced DBCarver tool for database forensic analysis.

S. No	Title	Author	Focus/Objective	Conclusion
9	Categorizati-on and organization of database forensic investigation processes(20 16)	Arafat Al-Dhaqm et al	ICICT Conference	Proposed **classification model for database forensic processes**.
10	Database Forensics Field and Children Crimes.(2017)	Al-Dhaqm, Arafat, et al.	PeerJ Computer Science	Kids Cybersecurity Using Computational Intelligence Techniques
11	Digital forensic analysis of attack detection.(2019)	Jyotinagar, Varshapriya , and Bandu Meshram	Journal of Integrated Science and Technology	Concludes that digital forensic techniques significantly improve attack detection and identification .
12	Enhancing Log File Analysis in Digital Forensics .(2019)	Natarajan, Arul Kumar, et al	Cyber Security, Forensics and National Security. CRC Press,	Demonstrates that machine learning enhances log file analysis accuracy and efficiency in digital forensics and incident response.

S. No	Title	Author	Focus/Objective	Conclusion
13	Digital forensics analysis of iot nodes using machine learning (2020)	Arshad, M. Zeeshan, et al	Journal of Computing & Biomedical Informatics	Shows that machine learning-based forensic analysis effectively detects and investigates security incidents in IoT nodes.
14	Digital forensics analysis of iot nodes using machine learning (2020)	Arshad, M. Zeeshan, et al	Journal of Computing & Biomedical Informatics 4.01: 1-12.	Reaffirms that integrating ML with IoT forensics improves evidence extraction and attack identification.
15	Web application attack detection and forensics: A survey. (2021)	Baker, Mohammed, Enis Karaarslan, and Yasar Hoscanbi	6th international symposium on digital forensic and security (ISDFS). Ieee.	Concludes that combining web attack detection mechanisms with forensic readiness strengthens web application security.
16	Evidence detection in cloud forensics (2021)	Abuowaida, Suhaila, et al.	Data and Metadata 4 : 699.	Establishes that ML-based classification models effectively detect cyber-attacks in IaaS cloud forensic investigations.

S. No	Title	Author	Focus/Objective	Conclusion
17	Forensic analysis and detection of spoofing based email attack using memory forensics and machine learning (2022)	Shukla, Sanjeev, Manoj Misra, and Gaurav Varshney.	Cham: Springer Nature Switzerland, 2022.	Concludes that integrating memory forensics with ML improves spoofed email attack detection accuracy.
18	Forensic analysis on internet of things (IoT) device using machine-to-machine (2022)	Mazhar, Muhammad Shoaib, et al	Electronics 11.7 (2022): 1126.	Shows that M2M-based forensic frameworks enhance investigation capabilities in IoT device environments..
19	Database security threats and challenges in database forensic:A survey.(2025)	Imtiaz, Muhammad Atif, et al	Spectrum of Engineering Sciences (2025): 272-303.	Explained database threats and forensic challenges
20	Digital Forensics as Advanced Ransomware Pre-Attack Detection Algorithm for Endpoint Data Protection (2025)	Du, Jian, et al	Security and Communication Networks 1424638.	Demonstrates that advanced digital forensic algorithms can detect ransomware threats before full-scale attacks occur.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing web authentication systems mainly depend on traditional security mechanisms such as input validation and firewall protection. These systems have several limitations:

- Manual rule-based filtering cannot detect advanced attack patterns.
- Lack of intelligent prediction mechanisms for unknown attacks.
- Absence of forensic logging for security investigation.
- Low efficiency in detecting SQL Injection and XSS payload variations.
- No real-time monitoring of suspicious login attempts.

Traditional systems are vulnerable to modern cyberattack techniques because attackers continuously modify payload structures to bypass security filters.

Disadvantages of the Existing System

The existing authentication and web security systems mainly rely on traditional rule-based validation techniques and lack intelligent threat detection mechanisms. The major disadvantages are listed below:

a. Lack of Intelligent Security Detection

Existing systems use simple input validation methods which cannot identify complex or evolving attack patterns such as advanced SQL Injection and Cross-Site Scripting (XSS) attacks.

b. No Machine Learning Integration

Traditional systems do not use machine learning algorithms for behavior-based or payload-based attack prediction, resulting in lower detection capability.

c. Poor Handling of Unknown Attacks

If attackers modify payload structures or use new attack variations, the system may fail to detect malicious activities.

d. Absence of Forensic Logging

Most existing systems do not maintain detailed logs of login attempts, IP addresses, timestamps, or attack classifications, making forensic investigation difficult.

3.2 PROPOSED SYSTEM

The proposed system integrates Machine Learning-based detection along with rule-based security.The system uses datasets containing normal and malicious payloads to train a classification model.

The login input is processed using TF-IDF vectorization, and Random Forest classification is applied to predict attack types.

Key features of the proposed system include:

- Intelligent detection of SQL Injection and XSS attacks.
- Secure login authentication mechanism.
- Database forensic logging system.
- IP address and timestamp tracking.
- Confidence-based prediction results.
- Command prompt monitoring.

Advantages of the Proposed System

1. Improves web application security.
2. Provides early attack detection.
3. Supports forensic investigation.
4. Reduces manual security monitoring.
5. Enhances accuracy using machine learning.

4. SYSTEM OVERVIEW

The **ML-Based Login Attack Detection and Database Forensics System** is designed to enhance web authentication security by using machine learning techniques to detect malicious login attempts. The system provides intelligent analysis of user input credentials and identifies potential cyberattacks such as SQL Injection and Cross-Site Scripting (XSS). The system consists of three main components: user interface, machine learning detection module, and database logging module. The user interface provides a secure and attractive login webpage where users can enter their username and password.

When a login attempt is made, the system processes the input data and analyzes it using a trained machine learning model. The input credentials are converted into numerical features using TF-IDF vectorization, and classification is performed using the Random Forest algorithm. The model predicts whether the input is normal or malicious and calculates the confidence level of detection.

In addition to machine learning-based detection, the system also implements rule-based filtering to improve security performance. Common malicious keywords and payload patterns are checked before applying ML classification.

The system also maintains forensic logs by storing login details such as username, IP address, timestamp, attack type, and confidence score in a MySQL database. The log data helps security administrators perform attack investigation and monitoring.

The system operates in real time and displays detection results in the command prompt for security observation. The main goal of the system is to provide intelligent login security.

4.1 Working Process of the System:

The working process of the **ML-Based Login Attack Detection and Database Forensics System** is designed to provide secure authentication and intelligent attack detection using machine learning and rule-based filtering.

a. User Login Input

The process begins when a user accesses the login webpage and enters their username and password. The system collects the input credentials for analysis

b. Input Preprocessing

The entered username and password are combined and processed using text preprocessing techniques. The system converts textual input into numerical feature representation using **TF-IDF (Term Frequency – Inverse Document Frequency)** vectorization.

c. Attack Detection Phase

The system uses a hybrid detection mechanism consisting of:

- **Rule-Based Filtering**
  - The system checks for common malicious keywords such as SQL Injection payloads and XSS script patterns.
  - If malicious keywords are detected, the system immediately classifies the input as an attack.
- **Machine Learning Classification**
  - If rule-based filtering does not detect an attack, the input is passed to the trained Machine Learning model.
  - The system uses the **Random Forest classification algorithm** to predict whether the input is normal or malicious.
  - The model also calculates confidence accuracy.

d. Database Logging

After detection, the system stores login attempt details in the MySQL database. The stored information includes:

- Username
- Hashed password
- IP address
- Timestamp
- Attack type (Normal, SQL Injection, or XSS)
- Confidence score

This logging mechanism supports database forensic analysis.

e. Output Display

The system displays the detection result on the webpage and command prompt. If suspicious

4.2 ADVANTAGES OF THE SYSTEM

Early Attack Detection :

The system detects SQL Injection and XSS attacks in real time before they affect the database.

Machine Learning Based Security:

Instead of only rule-based detection, it uses an ML model to identify suspicious patterns more intelligently.

Real-Time Monitoring :

Login attempts are analyzed instantly and attack type with confidence score is displayed in the command prompt.

Forensic Logging :

All login activities (username, password, IP address, date, time, attack type, status) are stored in MySQL for investigation.

Improved Authentication Security :

Protects login pages from common web injection attacks.

Attack Classification :

Clearly categorizes attempts as Normal, SQL Injection, XSS, or Suspicious**.**

Evidence Collection :

Stored logs help in digital forensic analysis and tracking attacker behavior.

Scalable and Extendable :

Can be upgraded to detect more types of cyberattacks in the future.

4.3 Challenges and Considerations

Challenges:

a. **Dataset Quality and Accuracy** – The performance of the ML model depends on the quality of SQL and XSS datasets. If the dataset contains noise or missing values, accuracy may reduce.

b. **False Positives and False Negatives** – The model may sometimes classify a normal login as suspicious (false positive) or miss a real attack (false negative).

c. **Model Training Complexity** – Proper preprocessing, vectorization, and model selection are required to achieve good detection performance.

d. **Real-Time Performance** – Analyzing every login request using ML must be fast enough to avoid delay in user authentication.

e. **Database Security** – Since logs store sensitive data like username, password, and IP address, proper database protection is necessary.

f. **Scalability Issues** – If the number of users increases, the system must handle large volumes of login attempts and logs efficiently.

Considerations:

a. **Data Cleaning and Preprocessing** – Always remove null values and convert text properly before training the ML model.

b. **Regular Model Updates** – The ML model should be retrained periodically with new attack datasets to handle evolving threats.

c. **Secure Storage** – Sensitive information should be encrypted in the database to maintain privacy.

d. **Performance Optimization** – Efficient algorithms and indexing in MySQL should be used to maintain fast processing.

5. METHODOLOGY

5.1 WORKFLOW

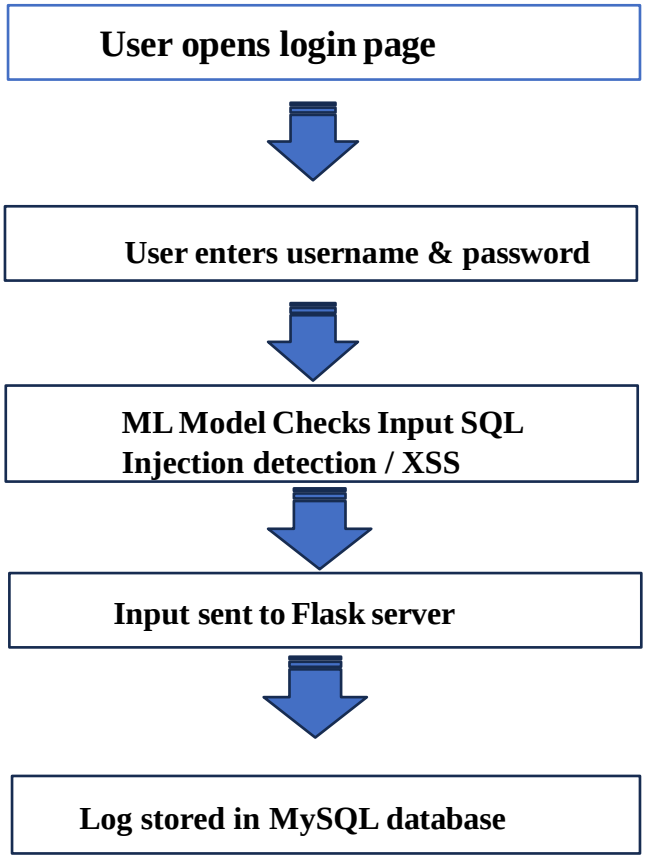

Fig 1. Workflow of ML-Based login attack detection and database forensics system.

5.2 TECHNOLOGY USED

1. **Frontend:** HTML, CSS, JavaScript (for attractive login website design)
2. **Backend:** Python Flask Framework
3. **Machine Learning:** Scikit-learn (Text Vectorization + Classification Model)
4. **Database:** MySQL 8.0 (Port 3307)

6. RESULT AND DISCUSSION

User accesses the secure login webpage built using Flask;It's shown as fig 2.

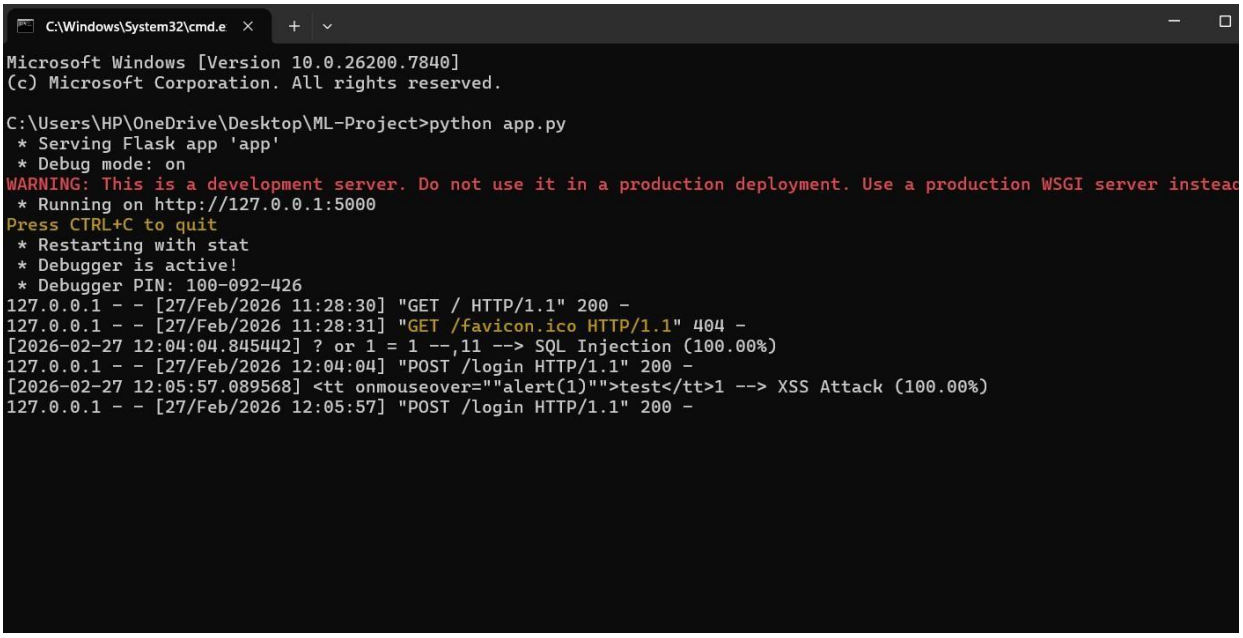

Fig 2. Flask application running locally on 127.0.0.1:5000

User enters login credentials in input fields;It's shown in fig 3.

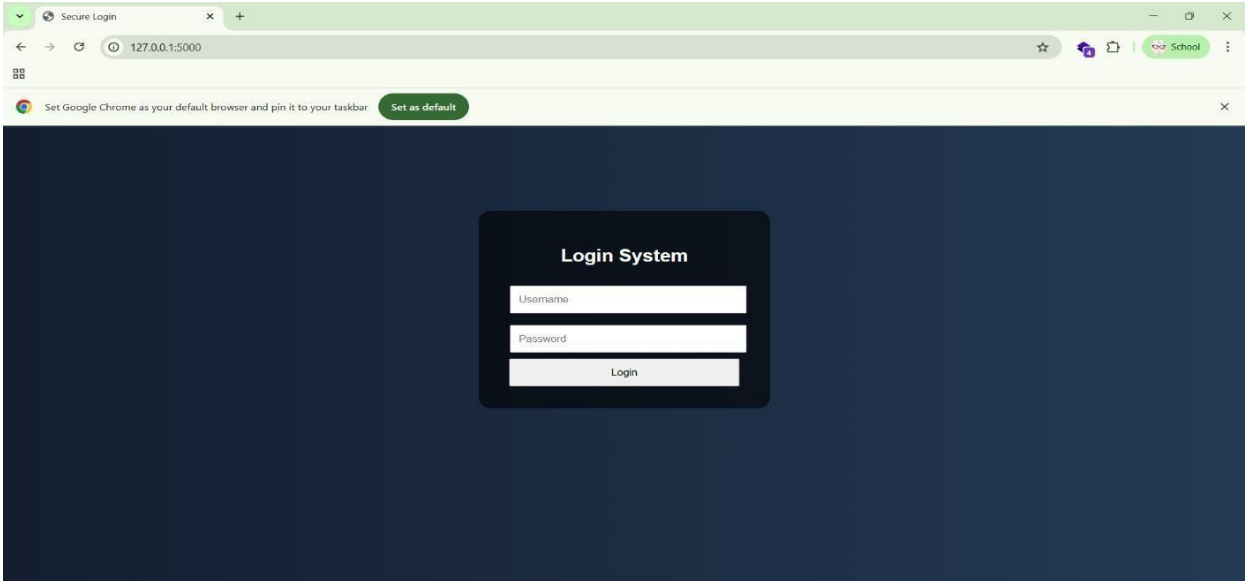

Fig 3. Enter username and password.

Username and password are combined and sent to the detection module;It’s shown in fig 4.

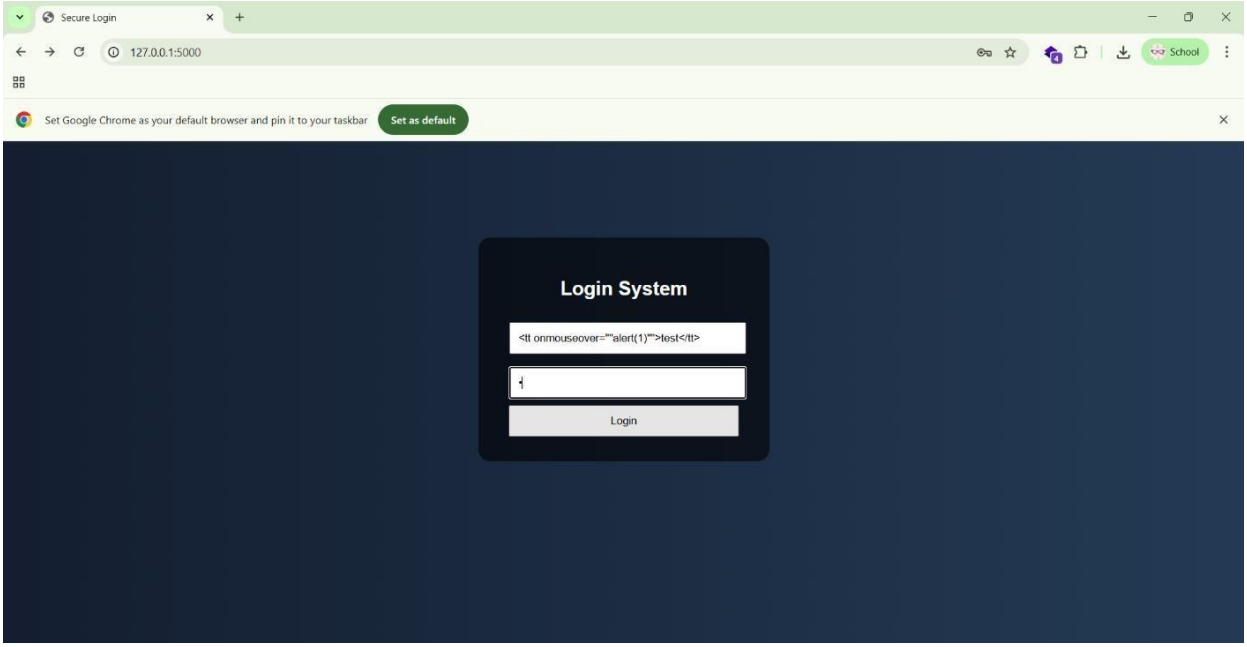

Fig 4. Cross-Site Scripting payload entered in username field for attack detection testing.

Input text is converted into numerical features using TF-IDF vectorization.
Random Forest classifier predicts whether input is Normal, SQL Injection, or XSS Attack;It’s shown in fig 5.

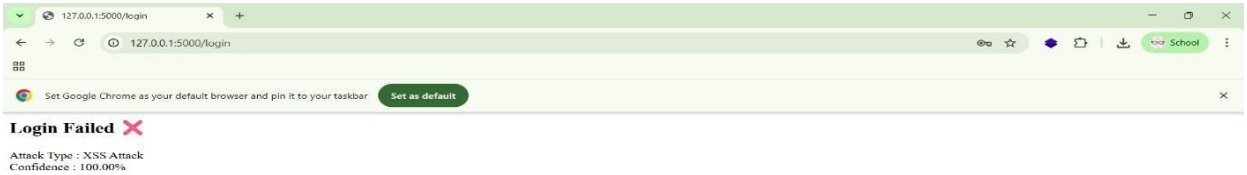

Fig 5. System identified malicious input and displays XSS attack detection .

Username and password are combined and sent to the detection module; It's shown in fig 6.

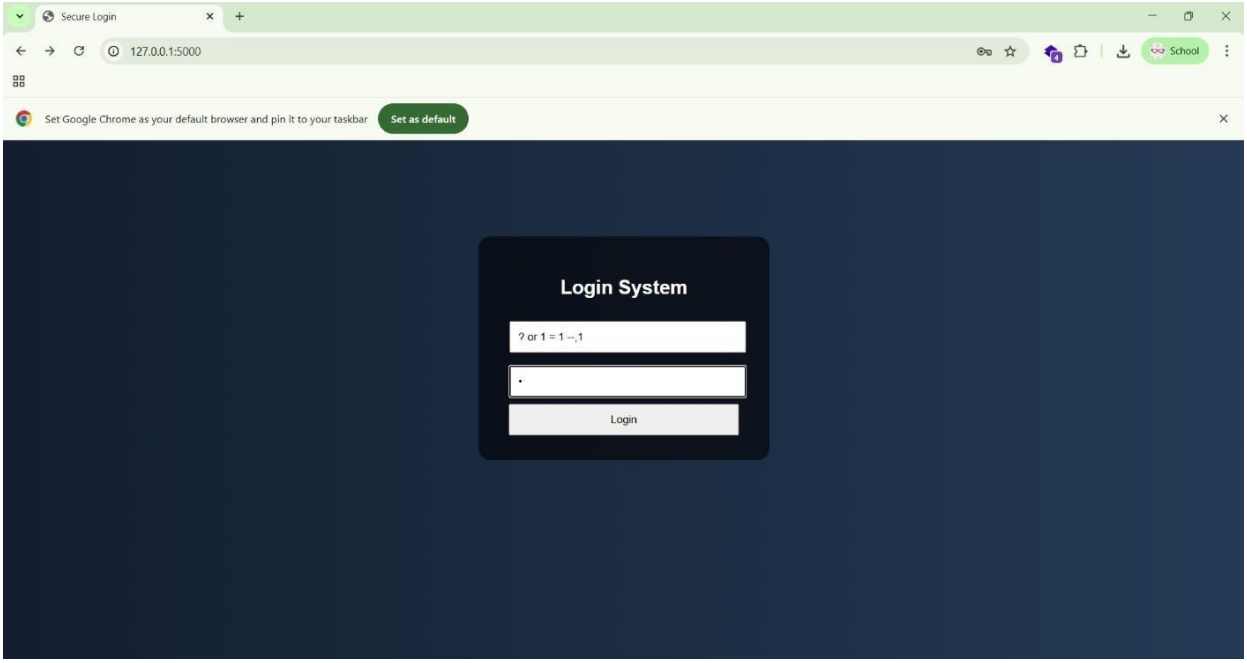

Fig 6. SQL Injection payload entered in username field for attack detection testing.

Input text is converted into numerical features using TF-IDF vectorization.Random Forest classifier predicts whether input is Normal, SQL Injection. It's shown in fig 7.

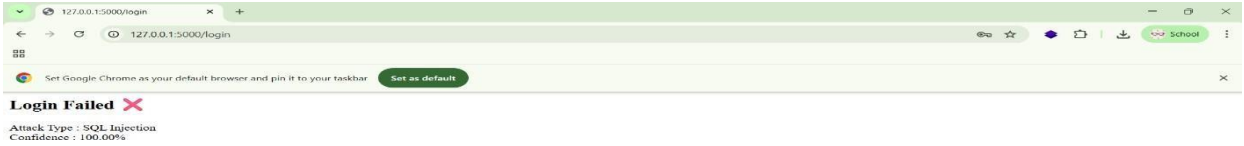

FIGURE6.6 System identified malicious input and displays SQL injection attack detection.

Login attempt details including username, IP address, timestamp, attack type, and confidence score are stored in MySQL database; It's shown in fig 8.

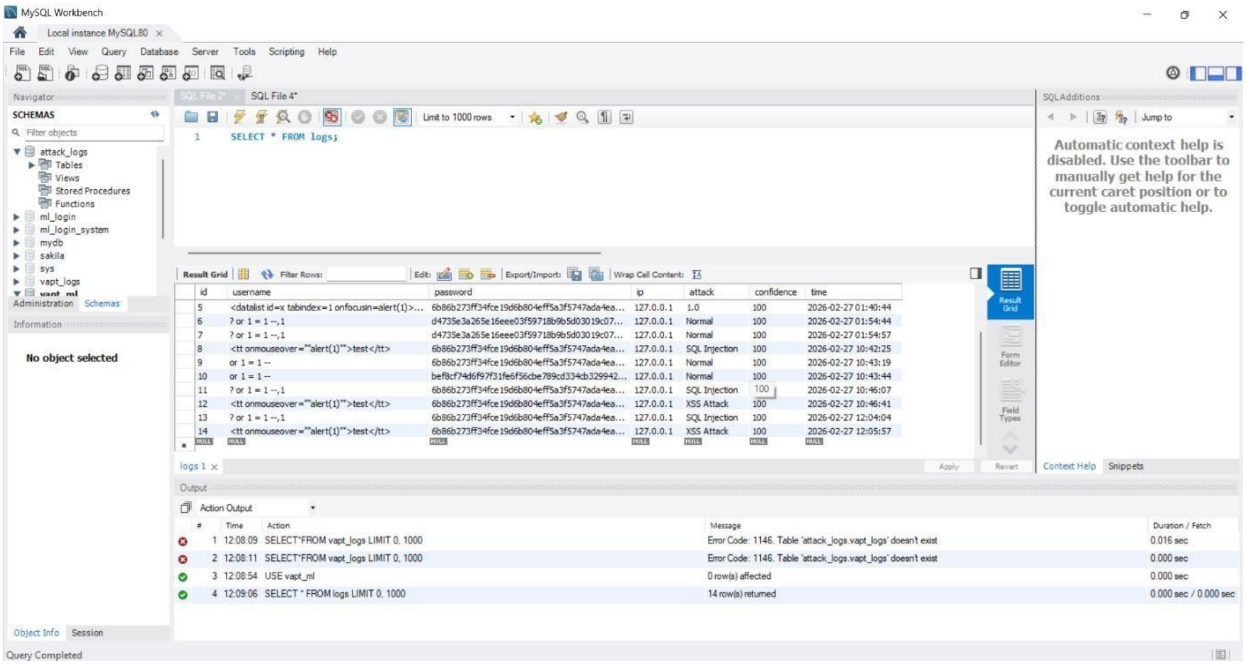

Fig 8. Stored logs in mysql database

7. FUTURE ENHANCEMENT

The future enhancement of this project can include improving the accuracy of attack detection by using advanced deep learning models such as neural networks or transformer-based classifiers.

The system can be extended to detect more types of web attacks such as brute force attacks, session hijacking, phishing URL detection, and malware-based login attempts. Real-time monitoring can be implemented by integrating live traffic analysis and alert notification systems such as email, SMS, or dashboard alerts.

The website interface can be made more attractive and responsive using modern frontend frameworks. Multi-factor authentication (MFA) can be added to provide stronger security protection. Additionally, the forensic module can be improved by generating detailed security reports, visualization charts, and automatic log anomaly detection. Cloud database integration and scalability optimization can also be considered for handling large-scale user authentication environments.

8. CONCLUSION

The ML-Based Login Attack Detection and Database Forensic System is developed to improve web authentication security by detecting malicious login attempts using Machine Learning techniques. In modern web applications, login pages are common targets for cyberattacks such as SQL Injection and Cross-Site Scripting (XSS). This project provides a secure login website where users can enter credentials, and the system will analyze the input using a trained ML model.

The model is trained using SQL injection and XSS payload datasets, where text vectorization and classification algorithms are used to detect suspicious patterns. When a user enters login information, the system classifies the attempt as normal or malicious and displays the result in the command prompt with confidence accuracy.

All login activities are stored in MySQL database 8.0 running on port 3307, including username, password, IP address, date, time, attack type, and login status. This forensic logging system helps security analysts track suspicious behaviour and investigate attacks. The project enhances cybersecurity by providing early detection of injection attacks and demonstrates how machine learning can be applied in authentication security systems.

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